

often experienced the ‘underwater voice effect’: this happens when you’re in an area with a poor signal, or if your current cell is crowded with traffic. Your phone can’t pick up all the time slots, resulting in the other person’s voice breaking.

A variation of the TDMA technology is Extended TDMA (ETDMA), developed by Hughes. The system here is dynamic; time slots are not assigned just because a call has been made, but only when there is data to be transferred. Consider this: you are on a call with your irate boss. Naturally, you aren’t doing much talking. Even so, your network has assigned a frequency and a time slot for you to transmit on. In the ETDMA network, your empty time slots will be used to transmit data from other calls.

The Global System For Mobile Communications - GSM

GSM is essentially a standard or a set of recommendations to set up TDMA-based mobile telephone networks. The advantage of setting a standard was that callers who subscribed to a GSM network would be able ‘roam’ outside their own home networks and into other GSM networks worldwide. Most of the world’s GSM networks operate on 900 MHz and 1800 MHz; major parts of the Americas, however, operate on 850 MHz and 1900 MHz. Each frequency band in a GSM network was divided into eight time-slots. GSM even introduced the concept of the *Subscriber Identity Module (SIM)* card, which stored your subscription information, the operator’s information, and had some memory space for you to store your phone book. This meant that switching to a new handset would be quite simple, and you’d still have all your contacts. GSM networks also offered text messaging at low costs in addition to support for voice, data, fax calls and e-mail.

Data over the GSM network is usually transferred using the *Circuit-Switched Data (CSD)* technique. While it was quite robust, it wasn’t very good for transferring large amounts of data. We’ll look at the difference between circuit-switched and packet-switched data later.